

05 FUNCTIONAL REQUIREMENTS

Functional Requirements

A concise system-level specification of what TRINETRA must be able to do.

Requirements register

ID	Module	System requirement
FR-01	Registration / Login	The system shall allow users to register and securely log in.
FR-02	Disaster Dashboard	The system shall display active disasters with type, location, severity, and status.
FR-03	Disaster Map	The system shall display disaster locations and affected areas on an interactive map.
FR-04	Safe Zones	The system shall display designated safe zones to users.
FR-05	Emergency Resources	The system shall display nearby hospitals, shelters, relief centers, and other emergency resources.
FR-06	Disaster Alerts	Authorized personnel shall be able to create and publish disaster alerts.
FR-07	Citizen Reporting	Users shall be able to report incidents with location, disaster type, description, and optional image.
FR-08	Authority Dashboard	Authorized personnel shall monitor disasters, reports, alerts, resources, and map context.
FR-09	Risk Assessment	The system shall calculate or display a risk level for a monitored location using available disaster data.
FR-10	AI/ML Analysis	Where applicable, the system shall process relevant data to generate bounded risk, classification, or recommendations.
FR-11	Resource Management	Authorized personnel shall be able to add and update emergency resources.
FR-12	Report Management	Authorities shall be able to view, validate, update, and manage citizen reports.

Cross-cutting acceptance rules

- Role and permission checks protect authority-only actions.
- Safety-critical information includes source, status, and freshness where available.
- Invalid input fails clearly without silently creating or changing records.
- Every service-dependent feature has an explicit unavailable or fallback state.
- AI output is advisory, versioned, and never presented as autonomous authority.